Data_viz

Practical task 1

1.1 The three countries with the lowest gender wage gap are:

1. Costa Rica
2. Belgium
3. Denmark

1.2 The three countries with the highest gender wage gap are:

1. South Korea
2. Japan
3. Chile

1.3. Costa Rica recorded the lowest gender wage gap among the OECD countries in 2015 due to a combination of strong labour policies, relatively high female participation in education, and legislation promoting equal employment opportunities. The country has laws that prohibit gender discrimination in the workplace and support equal pay for equal work. Women in Costa Rica have high levels of educational attainment, enabling them to compete for skilled jobs with better salaries. Public investment in education and healthcare has also helped increase women's participation in the labour market. Although gender inequality still exists in some sectors, Costa Rica's commitment to labour rights, social protection, and greater access to employment opportunities has contributed significantly to reducing the wage gap compared with many other OECD countries.

2.1 During March 2020, all three lines on the graph show a sharp increase in the price of isopropanol, indicating a sudden surge in demand. Prices remained relatively stable from May 2019 to February 2020 but rose dramatically in March 2020. This suggests that isopropanol became significantly more valuable due to increased market demand and limited supply. The graph clearly shows an unusual spike compared with the previous months, reflecting a major change in the market.

2.2 The sharp increase in isopropanol prices during March 2020 was most likely caused by the COVID-19 pandemic. As the virus spread globally, demand for hand sanitizer and disinfectants increased dramatically because isopropanol is a key ingredient in these products. Consumers, businesses, and healthcare facilities purchased large quantities of sanitizer, leading to shortages of isopropanol. The sudden increase in demand, combined with limited production capacity, caused prices to rise significantly during this period.

3. The bubble plot shows a positive relationship between GDP per capita and CO_2 emissions per person. In general, countries with higher incomes tend to produce more CO_2 per person because of greater industrial activity, energy consumption, and transportation use. However, the strength of this relationship differs across continents.

- Most African countries appear in the lower-left section of the graph, indicating both low GDP per capita and low CO_2 emissions per person. This suggests lower levels of industrialization and energy consumption compared with other continents.
- Asian countries are widely spread across the graph. Many have relatively low to moderate GDP and CO_2 emissions, while some wealthier countries have much higher emissions. The largest bubbles, representing countries such as China and India, indicate very large populations, although their emissions per person vary considerably.
- European countries generally have high GDP per capita with moderate to high CO_2 emissions. While they are economically developed, many countries have adopted cleaner technologies and environmental policies, helping to limit emissions compared with some other wealthy regions.
- North American countries tend to have high GDP per capita and high CO_2 emissions per person. Their economies are highly industrialized, with high energy use and widespread reliance on private transport, contributing to greater emissions.
- Most South American countries fall in the middle of the graph, showing moderate GDP per capita and moderate CO_2 emissions. Their emissions are generally lower than those of North America and Europe but higher than many African countries.
- Oceania includes countries with relatively high GDP per capita and high CO_2 emissions per person. Although the population sizes are generally smaller, economic development and resource-based industries contribute to higher emissions.

Therefore, the graph indicates that higher GDP per capita is generally associated with higher CO_2 emissions per person, although there are exceptions. Differences in energy sources, industrial structure, environmental policies, and technological development influence how much CO_2 countries emit at similar income levels.

Practical tasks 2

1. The graphs along the diagonal are histograms. They show the distribution (frequency) of each individual variable, namely:

   - Size of Living Area
   - Size of Garage
   - Year House was Built
   - Sale Price

2. Most garages are smaller than 1000 ft^2**.**

   The histogram for Size of Garage shows that most garages are concentrated between approximately 200 ft^2 and 800 ft^2, with relatively few garages exceeding 1000 ft^2**.**

3. The most expensive houses were generally built after 1950**.**

   The scatterplot comparing Year House was Built and Sale Price shows that higher-priced homes are mostly concentrated among newer houses, particularly those built from the 1950s onward, with many of the highest-priced homes built after 2000.

4. There is a strong positive relationship between Size of Living Area and Sale Price.

   As the size of the living area increases, the sale price generally increases as well. Although there is some variation, larger homes consistently tend to sell for higher prices, indicating a strong positive correlation between these two variables.